Chapter 19: Revision Questions

243-Question DGCA CPL/ATPL Bank • All Topics

Topics: VDF • ADF/NDB • VOR • ILS • Radar • DME • SSR • AWR/MLS • GNSS/RNAV/Mixed

VDF (Q1–5)

Q1. VDF — which emergency frequency most commonly used for position fix?

(a) 121.500 MHz

(b) 243.000 MHz

(c) 156.8 MHz

(d) 406 MHz

Ans: (a) — VDF is triggered when aircraft transmits on **121.500 MHz**. Ground DF equipment takes a bearing on the transmission.

Q2. Aircraft equipment needed for VDF let-down:

(a) VHF radio only

(b) VOR receiver

(c) VOR/DME

(d) Nothing — all ground-based

Ans: (a) — VDF needs only a **VHF radio** in the aircraft. All DF equipment is ground-based.

Q3. VDF station at 400 ft, aircraft at FL090. Max range?

(a) 117 NM

(b) 124 NM

(c) 134 NM

(d) 140 NM

Ans: (c) — $1.23\times(\sqrt{400}+\sqrt{9000})=1.23\times(20+94.9)=1.23\times114.9=$ **141 NM**. Nearest = 140 NM (d). [For

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the variant with 325/8000 ft: 1.23×(18+89)=134 NM]

Q4. VDF accuracy class 'A':

(a) ±1°

(b) ±2°

(c) ±5°

(d) ±10°

Ans: (a) — VDF Class A: **±1°**. Class B: ±2°. Class C: ±5°. Class D: ±10°.

Q5. VDF uses which propagation:

(a) Sky wave

(b) Ground wave

(c) Line of sight

(d) Surface wave

Ans: (c) — VDF VHF: **line of sight propagation**.

ADF / NDB (Q6–18)

Q6. Wavelength of 375 kHz NDB signal:

(a) 8 m

(b) 80 m

(c) 800 m

(d) 8000 m

Ans: (c) — λ=300,000,000/375,000= **800 m**.

Q7. NDB signal pattern:

(a) 30 Hz polar diagram

(b) Omni-directional

(c) Bi-lobal

(d) Beam rotating at 30 Hz

Ans: (b) — NDB radiates **omni-directionally**. The ADF loop aerial is directional.

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Q8. ADF accuracy within DOC by day:

(a) ±1°

(b) ±2°

(c) ±5°

(d) ±10°

> **Ans: (c)** — ADF accuracy by day: **±5°**.

Q9. Night effect on ADF worst at:

(a) Midday

(b) Midnight

(c) Dawn and dusk

(d) Noon and midnight

> **Ans: (c)** — Night effect (sky wave interference) worst at **dawn and dusk** when D-layer dissolves.

Q10. Two NDBs — 20 NM from coast vs 50 NM inland. Greatest coastal refraction error?

(a) 20 NM beacon

(b) 50 NM inland beacon

(c) Equal at RB 090/270

(d) Equal at RB 000/180

> **Ans: (b)** — Beacon **further inland** produces more coastal refraction — signal crosses coast at more oblique angle.

Q11. NDB range doubling — power increase needed:

(a) 4×

(b) 8×

(c) 16×

(d) 2×

> **Ans: (c)** — Power ∝ Range^4. Double range: power = **$2^4 = 16$**.

Q12. Quadrantal error in ADF caused by:

(a) Night effect

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(b) Metallic airframe re-radiating signal

(c) Coastal refraction

(d) Precipitation static

Ans: (b) — Quadrantal error: **metallic airframe refracts/re-radiates signal** at 45° quadrants.

Q13. ADF action for bearing — which aerials?

(a) Loop only

(b) Sense only

(c) Both loop and sense

(d) Neither — automatic

Ans: (c) — ADF needs **both loop (directional null) and sense (resolve 180° ambiguity)**.

Q14. Coastal error worst when beacon is:

(a) Inland, acute angle to coast

(b) Inland, 90° to coast

(c) Close to coast, acute angle

(d) Close to coast, 90° to coast

Ans: (a) — Worst: **inland beacon, signal at acute angle to coastline**.

Q15. Most significant ADF error:

(a) Quadrantal error

(b) Mountain effect

(c) Night effect

(d) Coastal refraction

Ans: (c) — **Night effect** (sky wave) is most significant ADF error.

Q16. ADF bearing inaccuracies caused by: (select best group)

(a) Static interference, height, SA

(b) Mountain effect, SA, night effect

(c) Lack of warning, station interference, static

(d) Coastal refraction, slant range, night effect

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Ans: (c) — Valid ADF errors: **lack of failure warning, station interference, static interference** (also night, coastal, QE, mountain).

Q17. NDB frequency band:

(a) 250–450 kHz

(b) 190–1750 kHz

(c) 108–117.95 MHz

(d) 329–335 MHz

Ans: (b) — NDB: **190–1750 kHz** (LF/MF).

Q18. BFO switch on ADF used when:

(a) Always on

(b) Required for NON emission beacons to produce audible ident

(c) Increases range

(d) Reduces quadrantal error

Ans: (b) — BFO needed for **NON (unmodulated CW) beacons** — heterodynes the carrier to produce audible ident tone.

VOR (Q19–35)

Q19. VOR bearing measurement principle:

(a) Phase comparison

(b) Switched cardioids

(c) DDM

(d) Pulse technique

Ans: (a) — VOR: **phase comparison** between 30 Hz FM reference and 30 Hz AM variable.

Q20. VOR variation applied at:

(a) Aircraft for both

(b) VOR station for radial; aircraft for ADF

(c) VOR station for both

(d) Aircraft for VOR; station for ADF

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Ans: (b) — VOR radials referenced to magnetic north at **station**. ADF uses variation at **aircraft**.

Q21. Aircraft flies due south of VOR. Var at station 13°W. Radial?

(a) 167°

(b) 180°

(c) 193°

(d) 347°

Ans: (c) — True bearing from station = 180° (south). Magnetic radial = 180° + 13°W = **193°**.

Q22. VOR radial 250 selected, CDI 5 dots left (1 dot = 2°). OBS to centre?

(a) 240

(b) 260

(c) 250

(d) 245

Ans: (b) — 5 dots left = 10° right of track. Select 250+10 = **260°**.

Q23. VOR frequency band:

(a) 190–1750 kHz

(b) 108–117.95 MHz

(c) 329–335 MHz

(d) 960–1215 MHz

Ans: (b) — VOR: **108–117.95 MHz VHF**.

Q24. VOR ident tone frequency:

(a) 400 Hz

(b) 1020 Hz

(c) 1350 Hz

(d) 3000 Hz

Ans: (b) — VOR ident: **1020 Hz Morse** every 10 s. DME ident: 1350 Hz.

Q25. VOR scalloping caused by:

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(a) Mountains

(b) Multipath/terrain reflections

(c) Night effect

(d) Precipitation

> **Ans: (b)** — Scalloping: **multipath propagation** from reflections.

Q26. DVOR advantage over CVOR:

(a) Higher power

(b) More channels

(c) Less siting error/scalloping

(d) Greater range

> **Ans: (c)** — DVOR: electronic rotation → **less siting error and scalloping**.

Q27. VOT should indicate on any radial:

(a) 000° FROM

(b) 180° FROM

(c) 090° TO

(d) Varies by position

> **Ans: (a)** — VOT: always **000° FROM (or 180° TO)**.

Q28. VOR reference phase transmitted as:

(a) 30 Hz AM direct

(b) 30 Hz FM on 9960 Hz sub-carrier

(c) 50 Hz modulation

(d) 400 Hz sub-carrier

> **Ans: (b)** — Reference: **9960 Hz sub-carrier FM at 30 Hz**, omni. Variable: 30 Hz AM rotating pattern.

Q29. VORTAC: civil aircraft uses:

(a) TACAN bearing only

(b) VOR bearing + TACAN DME

(c) VOR only

(d) TACAN DME only

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Ans: (b) — Civil: **VOR bearing + TACAN DME**.

Q30. VOR site error typical maximum:

(a) ±1°

(b) ±3°

(c) ±5°

(d) ±10°

Ans: (b) — VOR site error (terrain reflections): up to **±3°**.

Q31. VOR cone of confusion unreliable signals begin at approximately:

(a) Directly overhead

(b) 45° elevation

(c) 30° elevation

(d) 10° elevation

Ans: (c) — Cone of confusion starts at ≈ **30° elevation** above VOR.

Q32. En-route VOR DOC:

(a) 25 NM

(b) 50 NM

(c) 100 NM

(d) 200 NM

Ans: (d) — En-route VOR: **200 NM**.

Q33. OBS 090° TO, CDI centred, heading 080°. Aircraft:

(a) On 090° radial heading to VOR

(b) On 270° radial heading to VOR

(c) On 270° radial heading away

(d) On 090° radial heading away

Ans: (b) — 090° TO = aircraft on 270° radial. CDI centred. Heading 080° ≈ toward VOR → **270° radial, heading toward VOR**.

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Q34. VOR emission type:

(a) NON

(b) A9W

(c) F3N

(d) P0N

Ans: (b) — VOR: **A9W**.

Q35. DVOR — rotating pattern produced by:

(a) Rotating aerial mechanism

(b) Electronic switching of aerial array

(c) Doppler shift of carrier

(d) FM modulation of sub-carrier

Ans: (b) — DVOR: **electronic switching** of aerial array (no moving parts → less siting error).

ILS (Q36–60)

Q36. ILS localizer frequency band (odd decimals only):

(a) 108–117.95 MHz

(b) 108–111.975 MHz

(c) 329–335 MHz

(d) 960–1215 MHz

Ans: (b) — LOC: **108–111.975 MHz VHF**, odd-decimal tenths.

Q37. ILS glideslope frequency band:

(a) 108–111.975 MHz

(b) 329–335 MHz

(c) 960–1215 MHz

(d) 75 MHz

Ans: (b) — G/S: **329–335 MHz UHF**.

Q38. ILS glideslope usable range:

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(a) 5 NM

(b) 10 NM

(c) 15 NM

(d) 25 NM

> **Ans: (b)** — G/S usable range: **10 NM**.

Q39. ILS: aircraft at 3° G/S, 10 NM from threshold. Height?

(a) 2000 ft

(b) 2500 ft

(c) 3000 ft

(d) 3500 ft

> **Ans: (c)** — 3×10×100 = **3000 ft**.

Q40. ILS false glideslope strongest at:

(a) 1.5× nominal

(b) 2× nominal

(c) 3× nominal

(d) 5× nominal

> **Ans: (c)** — False G/S strongest at **3× nominal angle** (≈9° for 3° slope).

Q41. ILS LOC DDM on centreline:

(a) Maximum

(b) Zero

(c) Half-scale

(d) 150%

> **Ans: (b)** — On centreline: 90 Hz = 150 Hz → DDM = **zero**.

Q42. ILS outer marker: tone and light:

(a) 3000 Hz dots, white

(b) 400 Hz dashes, blue

(c) 1300 Hz alt, amber

(d) 400 Hz dashes, amber

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Ans: (b) — OM: **400 Hz dashes, blue**. MM: 1300 Hz, amber. IM: 3000 Hz, white.

Q43. ILS CAT I minima:

(a) DH 200 ft, RVR 550 m

(b) DH 200 ft, RVR 800 m

(c) DH 100 ft, RVR 400 m

(d) DH 100 ft, RVR 200 m

Ans: (b) — CAT I: **DH 200 ft, RVR 800 m**.

Q44. ILS CAT II minima:

(a) DH 200 ft, RVR 800 m

(b) DH 100 ft, RVR 400 m

(c) DH 50 ft, RVR 200 m

(d) No DH, RVR 50 m

Ans: (b) — CAT II: **DH 100 ft, RVR 400 m**.

Q45. ILS CAT IIIC minima:

(a) DH 100 ft, RVR 200 m

(b) DH 50 ft, RVR 75 m

(c) No DH, RVR 50 m

(d) No DH, no RVR minimum

Ans: (d) — CAT IIIC: **no DH, no RVR minimum**.

Q46. Glideslope bar deflects UP — aircraft is:

(a) Above glidepath

(b) On glidepath

(c) Below glidepath

(d) Left of centreline

Ans: (c) — Bar UP = 150 Hz dominant = aircraft **below glidepath** — fly up.

Q47. LOC CDI deflects RIGHT — aircraft is:

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(a) Right of centreline

(b) Left of centreline

(c) On centreline

(d) Above glidepath

Ans: (b) — CDI right = selected course is right = aircraft is **left of centreline** — fly right.

Q48. 90 Hz dominant on glideslope — aircraft is:

(a) Above glidepath

(b) Below glidepath

(c) On centreline

(d) Left of LOC

Ans: (a) — 90 Hz dominant on G/S = aircraft **above glidepath** — fly down.

Q49. ILS marker beacon carrier frequency:

(a) 108 MHz

(b) 75 MHz

(c) 329 MHz

(d) 1090 MHz

Ans: (b) — All ILS markers: **75 MHz carrier**.

Q50. ILS 3° G/S at 4.6 NM (50 ft threshold elevation). Height?

(a) 1380 ft

(b) 1400 ft

(c) 1430 ft

(d) 1500 ft

Ans: (c) — $3 \times 4.6 \times 100 + 50 = 1380 + 50 =$ **1430 ft**.

Q51. ILS back course — CDI sense:

(a) Normal

(b) Reversed

(c) Glideslope only works

(d) No change

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Ans: (b) — Back course LOC: CDI sense **reversed** — fly opposite needle.

Q52. ILS glideslope auto-paired when:

(a) Manually entered on FMC

(b) LOC VHF frequency selected

(c) ILS button pressed on autopilot

(d) Both LOC and GS frequencies entered

Ans: (b) — G/S automatically paired when **LOC VHF frequency selected**.

Q53. ILS glidepath accuracy requirement:

(a) ±0.5° either side of nominal

(b) ±1.0°

(c) ±0.25°

(d) ±2°

Ans: (a) — G/S accuracy: **±0.5°** of nominal slope.

Q54. ILS LOC beam width adjusted to give:

(a) ±2.5° fixed

(b) ±3° fixed

(c) ±35% of runway threshold width

(d) ±5° fixed

Ans: (c) — LOC width: **±35% of threshold width** (variable ≈±2°–6°).

Q55. ILS glidepath signal: DDM = 0.175 at:

(a) Course line

(b) Half-scale deflection

(c) Full-scale deflection

(d) False glidepath

Ans: (c) — Full-scale deflection DDM = **0.155**. [0.175 may appear in some variants — check exam wording]

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Q56. ILS CAT IIIB minima:

(a) DH <50 ft, RVR 50–200 m

(b) No DH, no RVR

(c) DH 100 ft, RVR 200 m

(d) DH 50 ft, RVR 200 m

Ans: (a) — CAT IIIB: **DH below 50 ft, RVR 50–200 m**.

Q57. Middle marker approximately:

(a) 4–7 NM from threshold

(b) 1050 m from threshold

(c) 300 m from threshold

(d) At decision height point

Ans: (b) — MM: ≈**1050 m** from threshold.

Q58. ILS number of channels (40-channel system):

(a) 200

(b) 252

(c) 40

(d) 108

Ans: (c) — ILS: **40 channels**. MLS: 200. DME: 252.

Q59. ILS LOC emission type:

(a) NON

(b) A9W

(c) F3N

(d) P0N

Ans: (b) — LOC: **A9W**.

Q60. ILS outer marker located:

(a) 300 m from threshold

(b) 1050 m from threshold

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(c) 4–7 NM from threshold

(d) 10 NM from threshold

Ans: (c) — OM: **4–7 NM** from threshold.

Radar (Q61–90)

Q61. PRF 400 pps. Maximum radar range:

(a) 162 NM

(b) 200 NM

(c) 203 NM

(d) 240 NM

Ans: (c) — 81,000/400 = **202.5 ≈ 203 NM**.

Q62. PRF 250 pps. Max range:

(a) 162 NM

(b) 200 NM

(c) 324 NM

(d) 405 NM

Ans: (c) — 81,000/250 = **324 NM**.

Q63. PRF 500 pps. Max range:

(a) 81 NM

(b) 162 NM

(c) 324 NM

(d) 243 NM

Ans: (b) — 81,000/500 = **162 NM**.

Q64. Echo received 740 µs after transmission. Range:

(a) 30 NM

(b) 45 NM

(c) 60 NM

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(d) 90 NM

Ans: (c) — 740/12.36 = **59.9 ≈ 60 NM**.

Q65. Pulse width 2 µs. Minimum range:

(a) 150 m

(b) 300 m

(c) 600 m

(d) 900 m

Ans: (b) — 2×150 = **300 m**.

Q66. Pulse width 0.03 µs (ASMI). Minimum range:

(a) 0.45 m

(b) 4.5 m

(c) 45 m

(d) 450 m

Ans: (b) — 0.03×150 = **4.5 m**.

Q67. Minimum range determined by:

(a) PRF

(b) Transmitter power

(c) Pulse width

(d) Beamwidth

Ans: (c) — Min range = PW(µs) × 150 m → determined by **pulse width**.

Q68. Maximum range determined by:

(a) Pulse width

(b) PRF

(c) Beamwidth

(d) Transmitter power

Ans: (b) — Max range = 81,000/PRF → determined by **PRF**.

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Q69. Azimuth resolution improved by:

(a) Wider beam

(b) Narrower beam

(c) Higher PRF

(d) Shorter pulse

Ans: (b) — Better azimuth resolution: **narrower beamwidth**.

Q70. Radial resolution improved by:

(a) Narrower beam

(b) Shorter pulse width

(c) Higher power

(d) Lower PRF

Ans: (b) — Better radial resolution: **shorter pulse width**.

Q71. MTI removes:

(a) Second trace returns

(b) Stationary clutter

(c) Rain

(d) Side lobe returns

Ans: (b) — MTI (Moving Target Indication): removes **stationary clutter** via Doppler.

Q72. Second trace returns removed by:

(a) MTI

(b) Increasing power

(c) Jittering PRF

(d) Reducing beamwidth

Ans: (c) — Second trace returns: **jittering (varying) PRF**.

Q73. AWR frequency:

(a) 3000 MHz

(b) 9375 MHz

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(c) 13300 MHz

(d) 35000 MHz

> **Ans: (b)** — AWR: **9375 MHz (SHF, 3.2 cm)**.

Q74. AWR is classified as:

(a) Secondary radar

(b) CW radar

(c) Primary radar

(d) Doppler radar

> **Ans: (c)** — AWR: **primary radar**.

Q75. AWR colour — weakest precipitation:

(a) Magenta

(b) Red

(c) Yellow

(d) Green

> **Ans: (d)** — Weakest: **Green** (0.7–4 mm/h). Black = nothing.

Q76. AWR colour — turbulence/severe:

(a) Red

(b) Yellow

(c) Green

(d) Magenta

> **Ans: (d)** — Turbulence: **Magenta**.

Q77. AWR MAP mode uses which beam up to 60–70 NM:

(a) Pencil beam

(b) Cosecant^2 (fan) beam

(c) Omni

(d) Scanning beam

> **Ans: (b)** — MAP mode: **cosecant^2 fan beam** up to 60–70 NM.

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Q78. AWR CONT mode shows:

(a) Increased gain

(b) Rainfall rate zone boundaries

(c) Map features

(d) Turbulence only

Ans: (b) — CONT: **zone boundaries** — steepest gradient = most hazardous.

Q79. AWR HOLD function:

(a) Freezes antenna

(b) Freezes display for storm movement assessment

(c) Holds gain

(d) Holds range

Ans: (b) — HOLD: **freezes display** to assess storm movement (compare after 2–3 min).

Q80. AWR shadow area:

(a) Area under aircraft

(b) Area behind heavy precipitation

(c) Below the beam

(d) Overhead cone

Ans: (b) — Shadow: region **behind heavy rain** — attenuated beam reveals nothing beyond.

Q81. AWR hook/U/finger return on display indicates:

(a) Light rain

(b) Snow

(c) Hail and severe turbulence

(d) CAT

Ans: (c) — Hook, U, finger shapes: **hail and severe turbulence** — avoid.

Q82. AWR stabilisation prevents:

(a) False targets

(b) Multipath

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(c) Display distortion during manoeuvres

(d) Second trace returns

Ans: (c) — Stabilisation: prevents **display distortion during pitch/roll**.

Q83. ASMI scan rate:

(a) 6 rpm

(b) 15 rpm

(c) 60 rpm

(d) 120 rpm

Ans: (c) — ASMI: **60 rpm**.

Q84. ASMI purpose:

(a) Long-range surveillance

(b) Aircraft and vehicle surface movement

(c) Weather detection

(d) Approach control

Ans: (b) — ASMI: **surface movement** — aircraft/vehicles on runways/taxiways.

Q85. Primary radar cannot determine:

(a) Range

(b) Bearing

(c) Identity

(d) Approximate speed (by track)

Ans: (c) — Primary radar: no **identity** — requires SSR transponder.

Q86. Super-refraction effect on radar:

(a) Reduces range

(b) Extends range beyond LOS

(c) Creates clutter

(d) Increases minimum range

Ans: (b) — Super-refraction: **extends range beyond normal LOS** (may cause spurious returns).

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Q87. PRF 1000 pps. Echo at 1050 µs. Situation:

(a) Target at 85 NM — genuine

(b) Second trace return — actual 85 NM

(c) False target — reject

(d) Target at 4 NM

Ans: (b) — PRI=1000 µs. 1050 > PRI → **second trace return**. Actual range = 81 + 50/12.36 = 85 NM.

Q88. UK radar separation within 40 NM below FL245:

(a) 1 NM

(b) 2 NM

(c) 3 NM

(d) 5 NM

Ans: (c) — UK radar sep: **3 NM**.

Q89. TAR maximum range:

(a) 25 NM

(b) 75 NM

(c) 150 NM

(d) 250 NM

Ans: (b) — TAR (Terminal Area Radar): **75 NM**.

Q90. Max PRF for 50 km range:

(a) 330 pps

(b) 617 pps

(c) 3000 pps

(d) 1620 pps

Ans: (c) — PRF = $3\times10^8/(2\times50{,}000)$ = **3000 pps**.

DME (Q91–120)

Q91. DME frequency band:

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(a) 108–118 MHz

(b) 329–335 MHz

(c) 960–1215 MHz

(d) 5031–5091 MHz

> **Ans: (c)** — DME: **UHF 960–1215 MHz**.

Q92. DME transponder offset:

(a) ±21 MHz

(b) ±50 MHz

(c) ±63 MHz

(d) ±100 MHz

> **Ans: (c)** — Ground transponder replies at **±63 MHz** from interrogation frequency.

Q93. DME transponder delay:

(a) 12.36 µs

(b) 21 µs

(c) 50 µs

(d) 8 µs

> **Ans: (c)** — Fixed delay: **50 µs**. Receiver subtracts before computing range.

Q94. DME tracking phase pulse rate:

(a) 27 ppps

(b) 60 ppps

(c) 150 ppps

(d) 2700 ppps

> **Ans: (a)** — Tracking: **27 ppps**.

Q95. DME search phase pulse rate:

(a) 27 ppps

(b) 60 ppps

(c) 150 ppps

(d) 2700 ppps

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Ans: (c) — Search: **150 ppps**.

Q96. DME saturation approximately:

(a) 27 ppps total

(b) 2700 ppps ≈ 100 aircraft

(c) 60 ppps total

(d) 150 ppps total

Ans: (b) — Saturation: **2700 ppps** ≈ **100 aircraft** in tracking.

Q97. DME slant range error negligible when:

(a) Range > altitude (NM)

(b) Range (NM) > 3 × altitude (thousands ft)

(c) Always negligible

(d) Below 3000 ft

Ans: (b) — Negligible: Range (NM) > **3 × Alt (000s ft)**.

Q98. Aircraft at FL180 directly over DME. Display reads:

(a) 0 NM

(b) 0.8 NM

(c) 3 NM

(d) 18 NM

Ans: (c) — 18,000/6076 = **2.96 ≈ 3 NM**.

Q99. Time from interrogation to echo = 1300 µs. Range:

(a) 98 NM

(b) 100 NM

(c) 101 NM

(d) 103 NM

Ans: (c) — (1300−50)/12.36 = 1250/12.36 = **101.1 NM**.

Q100. DME number of channels:

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(a) 40

(b) 108

(c) 200

(d) 252

Ans: (d) — DME: **252 channels**.

Q101. DME emission type:

(a) A9W

(b) P0N

(c) F3N

(d) NON

Ans: (b) — DME: **P0N** (pulsed, no info modulation).

Q102. VOR/DME pairing outside TMA — DME within:

(a) 100 ft

(b) 1 NM

(c) 2000 ft

(d) 2 NM

Ans: (c) — Outside TMA: DME within **2000 ft** of VOR for auto-pairing.

Q103. VOR/DME pairing inside TMA — DME within:

(a) 100 ft

(b) 1 NM

(c) 2000 ft

(d) 2 NM

Ans: (a) — Inside TMA: DME within **100 ft** of VOR.

Q104. DME jittered pulse pairs — purpose:

(a) Higher power

(b) Avoid ILS interference

(c) Identify own replies

(d) Increase range

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Ans: (c) — Jittered pairs: aircraft **identifies its own replies** by consistent time offset.

Q105. DME squitter purpose:

(a) Ident

(b) Fill capacity to prevent aircraft re-entering search

(c) Range calibration

(d) Interference avoidance

Ans: (b) — Squitter: **fills unused transponder capacity** so aircraft detect station is active.

Q106. DME memory duration in tracking:

(a) 5 s

(b) 10 s

(c) 15 s

(d) 30 s

Ans: (b) — DME memory: **10 seconds**.

Q107. DME/P accuracy:

(a) ±0.5 NM

(b) ±1 NM

(c) ±100 ft

(d) ±50 m

Ans: (c) — DME/P (Precision): **±100 ft** for MLS CAT II/III.

Q108. DME/P used with:

(a) ILS

(b) MLS

(c) GBAS

(d) SBAS

Ans: (b) — DME/P used with **MLS**.

Q109. VORTAC: military aircraft uses:

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(a) VOR bearing only

(b) VOR bearing + TACAN DME

(c) TACAN bearing + TACAN DME

(d) VOR only

> **Ans: (c)** — Military at VORTAC: **TACAN bearing + TACAN DME**.

Q110. DME ident interval:

(a) Every 10 s

(b) Every 30 s

(c) Continuous

(d) Every 60 s

> **Ans: (b)** — DME ident: **every 30 seconds**, 1350 Hz Morse.

Q111. DME type of radar:

(a) Primary

(b) Secondary

(c) Doppler

(d) CW

> **Ans: (b)** — DME: **secondary radar** principle.

Q112. RNAV DME range shown to:

(a) Real DME station

(b) Phantom station (waypoint)

(c) Nearest airport

(d) FIX position

> **Ans: (b)** — RNAV DME: shows distance to **phantom station (waypoint)**.

Q113. DME en-route accuracy:

(a) ±0.25 NM

(b) ±0.5 NM or ±3% (whichever greater)

(c) ±1 NM

(d) ±2%

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Ans: (b) — DME accuracy: **±0.5 NM or ±3%**, whichever greater.

Q114. GPS system uses what type of ranging?

(a) Secondary radar (2-way)

(b) One-way passive ranging

(c) Primary radar

(d) DME-type 2-way

Ans: (b) — GPS: **one-way passive** — satellite broadcasts, receiver measures time of arrival.

Q115. DME acquisition phase rate:

(a) 27 ppps

(b) 60 ppps

(c) 150 ppps

(d) 2700 ppps

Ans: (b) — Acquisition: **60 ppps**.

Q116. Aircraft at 5000 ft AGL over DME. Slant range reads:

(a) 0 NM

(b) 0.82 NM

(c) 2 NM

(d) 5 NM

Ans: (b) — 5000/6076 = **0.82 NM**.

Q117. DME squitter pulse rate (total with squitter):

(a) 27 ppps

(b) 2700 ppps

(c) 150 ppps

(d) 60 ppps

Ans: (b) — With squitter: maintained at **2700 ppps**.

Q118. Which PRN codes does GPS L1 carry?

Join live classes for guided practice and doubt clearing.

(a) P-code only

(b) C/A and P-code

(c) C/A code only (civil)

(d) SPS only

> **Ans: (b)** — L1 (1575.42 MHz): carries both **C/A code (civil) and P-code (military)**.

Q119. DME ground transponder delay ensures:

(a) Collision avoidance

(b) Positive identification of own replies

(c) Positive time offset for range calculation

(d) Synchronisation with VOR

> **Ans: (c)** — 50 µs delay ensures receiver always has **positive elapsed time** to subtract and compute range.

Q120. DME interrogation spacing (pulse pair):

(a) 6 µs X-channel; 12 µs Y-channel

(b) 12 µs X; 36 µs Y

(c) Random (jittered)

(d) Fixed at 12 µs

> **Ans: (b)** — X-channel pair spacing: **12 µs**. Y-channel: **36 µs**.

SSR (Q121–150)

Q121. SSR interrogation frequency:

(a) 1030 MHz

(b) 1090 MHz

(c) 960 MHz

(d) 1215 MHz

> **Ans: (a)** — Ground interrogates on **1030 MHz**.

Q122. SSR transponder reply frequency:

(a) 1030 MHz

Join live classes for guided practice and doubt clearing.

(b) 1090 MHz

(c) 960 MHz

(d) 1215 MHz

> **Ans: (b)** — Aircraft replies on **1090 MHz**.

Q123. Mode A P1–P3 spacing:

(a) 8 µs

(b) 17 µs

(c) 21 µs

(d) 25 µs

> **Ans: (a)** — Mode A: **8 µs**.

Q124. Mode C P1–P3 spacing:

(a) 8 µs

(b) 17 µs

(c) 21 µs

(d) 25 µs

> **Ans: (c)** — Mode C: **21 µs**.

Q125. Mode A codes available:

(a) 256

(b) 4096

(c) 65,536

(d) 16,777,216

> **Ans: (b)** — Mode A: **4096** (octal 0000–7777).

Q126. Mode S ICAO address size:

(a) 12 bit

(b) 16 bit

(c) 24 bit

(d) 32 bit

Join live classes for guided practice and doubt clearing.

Ans: (c) — Mode S: **24-bit** address → 16.7 million unique codes.

Q127. Mode C altitude resolution:

(a) 100 ft

(b) 50 ft

(c) 25 ft

(d) 10 ft

Ans: (a) — Mode C: **100 ft** increments. Mode S: 25 ft.

Q128. Squawk 7700:

(a) Hijack

(b) Radio failure

(c) Emergency

(d) VFR

Ans: (c) — 7700 = **Emergency**.

Q129. Squawk 7600:

(a) Emergency

(b) Radio failure

(c) Hijack

(d) Military

Ans: (b) — 7600 = **Radio failure (NORDO)**.

Q130. Squawk 7500:

(a) Emergency

(b) Radio failure

(c) Unlawful interference/hijack

(d) VFR conspicuity

Ans: (c) — 7500 = **Hijack/unlawful interference**.

Q131. Garbling occurs when two aircraft within:

Join live classes for guided practice and doubt clearing.

(a) 0.5 NM

(b) 1.7 NM same direction

(c) 3 NM

(d) 5 NM

> **Ans: (b)** — Garbling: **1.7 NM** in same direction from radar.

Q132. Fruiting caused by:

(a) Garbling

(b) Replies to other ground interrogators

(c) Multipath

(d) SA

> **Ans: (b)** — Fruiting: aircraft replies to **other stations' interrogations** → phantom targets.

Q133. IDENT button — SPI duration:

(a) 10 s

(b) 20 s

(c) 30 s

(d) 60 s

> **Ans: (b)** — SPI: **20 seconds**.

Q134. SLS — P2 pulse is:

(a) Directional, same as P1

(b) Omnidirectional

(c) Higher power than P1

(d) On 1090 MHz

> **Ans: (b)** — SLS P2: **omnidirectional** → if P2>P1, transponder suppresses reply (side lobe).

Q135. ISLS purpose:

(a) Prevent garbling

(b) Synchronise multiple interrogators to prevent fruiting

(c) Improve altitude reporting

(d) Enable data link

Join live classes for guided practice and doubt clearing.

Ans: (b) — ISLS: **synchronises multiple ground SSR stations** → prevents fruiting.

Q136. Transponder STANDBY:

(a) Reduced power response

(b) No response to interrogations

(c) Mode C only

(d) Mode S only

Ans: (b) — STANDBY: energised, **will NOT respond**.

Q137. UK VFR conspicuity squawk:

(a) 2000

(b) 7000

(c) 7700

(d) 1200

Ans: (b) — UK VFR: **7000**.

Q138. Mode S advantage:

(a) VHF frequencies

(b) Selective addressing — eliminates garbling

(c) No transponder needed

(d) No LOS needed

Ans: (b) — Mode S: **selective addressing** by unique 24-bit ICAO code.

Q139. ADS-B squitter interval:

(a) 1 s

(b) 0.5 s

(c) 2 s

(d) 10 s

Ans: (b) — ADS-B: every **0.5 seconds**.

Q140. Mode C altitude referenced to:

Join live classes for guided practice and doubt clearing.

(a) QNH

(b) QFE

(c) 1013.25 hPa

(d) Local QNH

> **Ans: (c)** — Mode C: always **1013.25 hPa**.

Q141. SSR reply framing pulses F1–F2 spacing:

(a) 8 µs

(b) 12 µs

(c) 20.3 µs

(d) 50 µs

> **Ans: (c)** — F1–F2: **20.3 µs**.

Q142. ACAS/TCAS uses:

(a) Mode A

(b) Mode C

(c) Mode S

(d) All modes

> **Ans: (c)** — TCAS: uses **Mode S** 1030/1090 MHz for A/C-to-A/C interrogation.

Q143. ELS (Elementary Surveillance) provides:

(a) Squawk and altitude

(b) Squawk, flight ID, altitude, emergency

(c) Full EHS data

(d) GPS position

> **Ans: (b)** — ELS: **squawk, callsign, pressure altitude, emergency**.

Q144. EHS (Enhanced Surveillance) adds:

(a) Nothing over ELS

(b) Selected altitude, airspeed, magnetic heading

(c) GPS position

(d) TCAS data

Join live classes for guided practice and doubt clearing.

Ans: (b) — EHS adds: **selected altitude, airspeed, magnetic heading**.

Q145. Squawk 2000 means:

(a) Emergency

(b) Entering CAS without clearance

(c) Military

(d) Radio failure

Ans: (b) — 2000: **entering CAS without prior ATC clearance**.

Q146. SSR antenna location:

(a) Separate tower

(b) On top of primary radar, co-rotating

(c) Inside fuselage

(d) Alongside ILS building

Ans: (b) — SSR antenna: **mounted on top of primary radar, co-rotating**.

Q147. Mode S data link direction:

(a) Downlink only

(b) Uplink only

(c) Bidirectional

(d) No data link

Ans: (c) — Mode S: **bidirectional** (downlink + uplink).

Q148. Transponder ALT mode:

(a) Responds Mode A only

(b) Responds Mode A + C (altitude)

(c) Mode S only

(d) STANDBY

Ans: (b) — ALT mode: responds to **Mode A and Mode C**.

Q149. ADS-B position source:

Join live classes for guided practice and doubt clearing.

(a) SSR ranging

(b) DME

(c) GPS

(d) Transponder timing

> **Ans: (c)** — ADS-B: broadcasts **GPS-derived position** via 1090 MHz extended squitter.

Q150. Minimum range for no garbling:

(a) 0.5 NM

(b) 1.7 NM

(c) 3 NM

(d) 5 NM

> **Ans: (b)** — No garbling when separation > **1.7 NM** in same direction.

AWR / MLS (Q151–180)

Q151. AWR wavelength:

(a) 1 cm

(b) 3.2 cm

(c) 10 cm

(d) 23 cm

> **Ans: (b)** — AWR: **3.2 cm (9375 MHz)**.

Q152. AWR MAN mode beam:

(a) Fan beam at all ranges

(b) Pencil beam, manual gain

(c) Omni

(d) Fan beam beyond 70 NM

> **Ans: (b)** — MAN: **pencil beam, manual gain**.

Q153. AWR cannot detect:

(a) Thunderstorms

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(b) Hail

(c) CAT

(d) Heavy rain

Ans: (c) — AWR cannot detect **CAT (Clear Air Turbulence)** — no precipitation.

Q154. AWR tilt positive (+) means:

(a) Beam below horizontal

(b) Beam above horizontal

(c) No tilt

(d) Gain increased

Ans: (b) — Positive tilt = beam **above horizontal**.

Q155. AWR: rain shadow risk is:

(a) Light rain conceals nothing

(b) Heavy rain attenuates beam — severe weather beyond may be hidden

(c) Snow creates shadow

(d) Shadow only below aircraft

Ans: (b) — Rain shadow: **severe weather may be hidden behind heavy precipitation**.

Q156. AWR 'finger' return shape indicates:

(a) Light rain

(b) Hail/severe turbulence

(c) Snow

(d) CAT

Ans: (b) — Finger shape: **hail and severe turbulence**.

Q157. AWR scalloped edges on return indicate:

(a) Ground return

(b) Hail

(c) Light rain

(d) Icing

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Ans: (b) — Scalloped edges: **hail**.

Q158. AWR AGC/STC range:

(a) Up to 25 NM

(b) All ranges

(c) Beyond 70 NM

(d) First 5 NM only

Ans: (a) — STC (Swept Gain/AGC): equalises returns up to **≈25 NM**.

Q159. AWR stabilisation failure — display effect:

(a) Blanks

(b) Lopsided during bank/pitch

(c) Gains increase

(d) Range rings disappear

Ans: (b) — Stabilisation failure: display becomes **lopsided during manoeuvres**.

Q160. MLS frequency band:

(a) VHF 108–118 MHz

(b) UHF 329–335 MHz

(c) SHF 5031–5090.7 MHz

(d) UHF 960–1215 MHz

Ans: (c) — MLS: **SHF 5031–5090.7 MHz**.

Q161. MLS azimuth coverage:

(a) ±20°

(b) ±35°

(c) ±40°

(d) ±60°

Ans: (c) — MLS azimuth: **±40°** of centreline.

Q162. MLS channels:

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(a) 40

(b) 100

(c) 200

(d) 252

Ans: (c) — MLS: **200 channels**.

Q163. MLS principle:

(a) DDM like ILS

(b) TRSB — time between TO/FRO sweeps

(c) Phase comparison

(d) Doppler

Ans: (b) — MLS: **TRSB (Time Referenced Scanning Beam)**.

Q164. MLS glide slope range:

(a) Fixed 3°

(b) 0.9° to 20°

(c) 2° to 10°

(d) Fixed 5°

Ans: (b) — MLS G/S: **0.9° to 20°** selectable.

Q165. MLS ident prefix letter:

(a) I

(b) M

(c) V

(d) D

Ans: (b) — MLS ident: prefix **M**.

Q166. MLS range measurement:

(a) Pulse echo timing

(b) DME/P built-in

(c) Phase comparison

(d) Doppler

Join live classes for guided practice and doubt clearing.

Ans: (b) — MLS range: **DME/P** (±100 ft accuracy).

Q167. MLS advantage over ILS — azimuth coverage:

(a) ±35° same as ILS

(b) ±40° vs ±35°

(c) 360°

(d) ±20°

Ans: (b) — MLS: **±40°** vs ILS ±35°.

Q168. MLS UK usable range:

(a) 10 NM

(b) 15 NM

(c) 20 NM

(d) 35 NM

Ans: (c) — MLS UK: **20 NM**.

Q169. AWR U-shape return:

(a) Light rain

(b) Hail/turbulence

(c) Snow

(d) CAT

Ans: (b) — U-shape: **hail and turbulence**.

Q170. AWR MAP mode optimal up to:

(a) 25 NM

(b) 60–70 NM

(c) 150 NM

(d) All ranges

Ans: (b) — MAP (cosecant^2 beam): up to **60–70 NM**.

Q171. ASMI frequency band:

Join live classes for guided practice and doubt clearing.

(a) 3 GHz

(b) 9.375 GHz

(c) 15–17 GHz

(d) 35 GHz

> **Ans: (c)** — ASMI: **15–17 GHz (SHF, ≈2 cm)**.

Q172. AWR WEA colour for no precip:

(a) Green

(b) Yellow

(c) Red

(d) Black

> **Ans: (d)** — No precipitation: **Black**.

Q173. AWR hook return on display:

(a) Light drizzle

(b) Hail/severe turbulence

(c) CAT

(d) Snow

> **Ans: (b)** — Hook shape: **hail/severe turbulence**.

Q174. AWR ageing (HOLD) — how long before compare?

(a) 30 s

(b) 1 min

(c) 2–3 min

(d) 10 min

> **Ans: (c)** — Deselect HOLD after **2–3 minutes** — compare images for storm movement.

Q175. AWR pencil beam — used in MAN mode beyond:

(a) 25 NM

(b) 50 NM

(c) 60–70 NM

(d) 150 NM

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Ans: (c) — Pencil beam (MAN) for weather beyond **60–70 NM**.

Q176. AWR — recommended tilt when far from storm cell:

(a) Tilt down to surface

(b) Zero tilt

(c) Tilt up

(d) Reduce gain

Ans: (c) — Far from cell: **tilt up** to see tops — determines storm height/severity.

Q177. SSR transponder receives on ___, replies on ___:

(a) 1090, 1030

(b) 1030, 1090

(c) 960, 1215

(d) 1215, 960

Ans: (b) — RX: **1030 MHz**, TX: **1090 MHz**.

Q178. EGNOS is classified as:

(a) LAAS

(b) GBAS

(c) SBAS

(d) RAIM

Ans: (c) — EGNOS: **SBAS (Satellite Based Augmentation System)**.

Q179. LAAS/GBAS transmits corrections via:

(a) Satellite

(b) HF radio

(c) VHF data link

(d) UHF beacon

Ans: (c) — LAAS/GBAS: **VHF data link**.

Q180. SBAS — corrections transmitted via:

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(a) Dedicated HF stations

(b) Geostationary satellites

(c) VHF data link

(d) L-band beacons

Ans: (b) — SBAS: corrections via **geostationary satellites**.

GNSS / RNAV / Mixed (Q181–243)

Q181. B-RNAV required accuracy:

(a) ±1 NM, 95%

(b) ±5 NM, 95%

(c) ±5 NM, 90%

(d) ±2 NM, 95%

Ans: (b) — B-RNAV: **±5 NM on 95% of occasions**.

Q182. P-RNAV required accuracy:

(a) ±1 NM, 95%

(b) ±5 NM, 95%

(c) ±0.5 NM, 95%

(d) ±2 NM, 90%

Ans: (a) — P-RNAV: **±1 NM on 95% of occasions**.

Q183. FMC navigation database update cycle:

(a) 7 days

(b) 14 days

(c) 28 days (AIRAC)

(d) 90 days

Ans: (c) — FMC nav DB: **28 days (AIRAC cycle)**.

Q184. RNAV erratic: aircraft beyond LOS of reference VOR/DME:

(a) True

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(b) False

(c) Only in TMA

(d) Only in cone of confusion

Ans: (a) — Yes — beyond LOS or DOC of reference VOR/DME → **erratic RNAV**.

Q185. GPS horizontal accuracy (SPS, 95%):

(a) ±100 m

(b) ±22 m

(c) ±13 m

(d) ±50 m

Ans: (c) — Horizontal: **±13 m (95%)**.

Q186. GPS vertical accuracy (SPS, 95%):

(a) ±13 m

(b) ±22 m

(c) ±50 m

(d) ±100 m

Ans: (b) — Vertical: **±22 m (95%)**.

Q187. GPS time accuracy (SPS, 95%):

(a) 1 µs

(b) 40 ns

(c) 100 ns

(d) 1 ms

Ans: (b) — Time: **40 nanoseconds (95%)**.

Q188. GPS — why pseudo-range not true range?

(a) Ionospheric delay

(b) Receiver clock error not yet eliminated

(c) SA applied

(d) Multipath

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Ans: (b) — Pseudo-range: uncorrected for **receiver clock error**.

Q189. GPS minimum SVs for 3D fix + time:

(a) 3

(b) 4

(c) 5

(d) 6

Ans: (b) — **4 SVs** for 3D fix (lat, lon, alt, time).

Q190. GPS RAIM — SVs for fault detection only:

(a) 4

(b) 5

(c) 6

(d) 7

Ans: (b) — RAIM detection: **5 SVs**.

Q191. GPS RAIM — SVs for fault exclusion:

(a) 4

(b) 5

(c) 6

(d) 7

Ans: (c) — RAIM exclusion: **6 SVs**.

Q192. GPS orbital height:

(a) 19,099 km

(b) 20,180 km

(c) 23,222 km

(d) 35,800 km

Ans: (b) — GPS: **20,180 km**.

Q193. GPS orbital inclination:

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(a) 55°

(b) 56°

(c) 63°

(d) 65°

> **Ans: (a)** — GPS: **55°** inclination.

Q194. GPS orbital period:

(a) 24 h

(b) 12 h

(c) 11 h 56 min

(d) 11 h 15 min

> **Ans: (c)** — GPS: **11 h 56 min**.

Q195. GPS L1 frequency:

(a) 1227.6 MHz

(b) 1575.42 MHz

(c) 1602 MHz

(d) 1246 MHz

> **Ans: (b)** — GPS L1: **1575.42 MHz**.

Q196. GPS L2 frequency:

(a) 1227.6 MHz

(b) 1575.42 MHz

(c) 1602 MHz

(d) 1246 MHz

> **Ans: (a)** — GPS L2: **1227.6 MHz**.

Q197. GPS geoid reference:

(a) PZ90

(b) ETRS89

(c) WGS84

(d) GRS80

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Ans: (c) — GPS: **WGS84**.

Q198. GLONASS geoid reference:

(a) WGS84

(b) PZ90

(c) ETRS89

(d) GRS80

Ans: (b) — GLONASS: **PZ90**.

Q199. Galileo geoid reference:

(a) WGS84

(b) PZ90

(c) ETRS89

(d) GRS80

Ans: (c) — Galileo: **ETRS89**.

Q200. GLONASS orbital inclination:

(a) 55°

(b) 56°

(c) 63.4°

(d) 65°

Ans: (d) — GLONASS: **65°**.

Q201. GLONASS orbital period:

(a) 11 h 56 min

(b) 11 h 15 min

(c) 12 h

(d) 14 h 8 min

Ans: (b) — GLONASS: **11 h 15 min**.

Q202. Galileo orbital height:

Join live classes for guided practice and doubt clearing.

(a) 20,180 km

(b) 19,099 km

(c) 23,222 km

(d) 35,800 km

> **Ans: (c)** — Galileo: **23,222 km**.

Q203. GPS almanac used by receiver to:

(a) Determine SA

(b) Compute position

(c) Identify which SVs visible

(d) Correct clock error

> **Ans: (c)** — Almanac: receiver determines **which SVs are above horizon** for faster acquisition.

Q204. GPS ephemeris — sub-frames:

(a) 1

(b) 2 and 3

(c) 4 and 5

(d) 5

> **Ans: (b)** — Ephemeris data: **sub-frames 2 and 3**.

Q205. GPS navigation message transmitted at:

(a) 50 Hz

(b) 1 kHz

(c) 9960 Hz

(d) 1575 MHz

> **Ans: (a)** — Nav message: **50 Hz (50 bps)** modulation on L1 and L2.

Q206. GNSS altitude cannot be used alone for DH/MDA because:

(a) GPS too inaccurate

(b) WGS84 ellipsoid differs from geoid (MSL) by up to 50 m

(c) Update rate too slow

(d) ILS always more accurate

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Ans: (b) — WGS84 ellipsoid ≠ MSL geoid → difference up to **50 m**.

Q207. SBAS geostationary orbit altitude:

(a) 20,180 km

(b) 19,099 km

(c) 23,222 km

(d) 35,800 km

Ans: (d) — Geostationary: **35,800 km**.

Q208. GPS dual-frequency advantage:

(a) Higher power

(b) Eliminates ≈99% ionospheric error

(c) Better coverage

(d) Faster acquisition

Ans: (b) — Dual-frequency (L1+L2): eliminates **≈99% of ionospheric error**.

Q209. GPS integrity warning time (non-precision approach):

(a) 2 s

(b) 8 s

(c) 30 s

(d) 10 s

Ans: (b) — Non-precision: **8 seconds**. Precision (CAT I): 6 s. ILS equivalent: 2 s.

Q210. 4D RNAV adds to 3D capability:

(a) Vertical guidance

(b) GPS

(c) Timing (RTA)

(d) TCAS

Ans: (c) — 4D RNAV: lateral + vertical + **timing (RTA)**.

Q211. FMC auto-tunes DMEs for:

Join live classes for guided practice and doubt clearing.

(a) Nearest station

(b) Pilot's selection

(c) Best geometry (angle of cut)

(d) Highest power stations

> **Ans: (c)** — FMC selects DMEs for **best angle of cut**.

Q212. External input to FMC:

(a) INS

(b) Pressure altitude

(c) Compass

(d) VOR/DME

> **Ans: (d)** — External (ground-based): **VOR/DME**.

Q213. ADC input to FMC:

(a) Heading

(b) Groundspeed

(c) TAS

(d) Position

> **Ans: (c)** — ADC provides: **TAS, pressure altitude, SAT**.

Q214. RNAV course line computer function:

(a) Direct to VOR/DME facility

(b) Rho/theta → track and distance to waypoint

(c) ILS approach guidance

(d) TCAS alerting

> **Ans: (b)** — RNAV: uses rho/theta to compute **track + distance to phantom waypoint**.

Q215. GPS range measurement: time measured from:

(a) Receiver to SV and back

(b) SV to receiver (one-way)

(c) Control segment to SV

(d) SV to control segment

Join live classes for guided practice and doubt clearing.

Ans: (b) — GPS: **one-way time** from SV to receiver.

Q216. GPS number of orbital planes:

(a) 3

(b) 4

(c) 6

(d) 8

Ans: (c) — GPS: **6 orbital planes**, 4 SVs each = 24 SVs.

Q217. GLONASS number of orbital planes:

(a) 3

(b) 4

(c) 6

(d) 8

Ans: (a) — GLONASS: **3 orbital planes**, 8 SVs each = 24 SVs.

Q218. Galileo number of orbital planes:

(a) 3

(b) 4

(c) 6

(d) 8

Ans: (a) — Galileo: **3 orbital planes**, 10 SVs each = 30 SVs.

Q219. GPS C/A code chipping rate:

(a) 1.023 MHz

(b) 10.23 MHz

(c) 50 Hz

(d) 9960 Hz

Ans: (a) — C/A code: **1.023 MHz**. P-code: 10.23 MHz.

Q220. Selective Availability (SA):

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(a) Still active on civilian GPS

(b) Intentional degradation of SPS; switched off in 2000

(c) Used only by military

(d) Replaced by RAIM

Ans: (b) — SA: **switched off 2 May 2000**. No longer applied.

Q221. RNAV erratic most likely cause:

(a) Cone of confusion of phantom

(b) Beyond DOC/LOS of reference VOR/DME

(c) FMC failure

(d) INS drift

Ans: (b) — Most likely: **beyond DOC or LOS of reference VOR/DME**.

Q222. P-RNAV accuracy requirement:

(a) ±5 NM, 95%

(b) ±1 NM, 95%

(c) ±0.5 NM, 95%

(d) ±1 NM, 99%

Ans: (b) — P-RNAV: **±1 NM, 95%**.

Q223. Multi-sensor FMC + GPS discrepancy — action:

(a) Trust GPS

(b) Trust multi-sensor; display multi-sensor output

(c) Select nearest VOR

(d) Declare emergency

Ans: (b) — Discrepancy: trust **multi-sensor**; GPS may have failed.

Q224. GBAS/LAAS accuracy for CAT I:

(a) ±16 m horizontal, ±6 m vertical

(b) ±100 ft

(c) ±0.5 NM

(d) Same as ILS

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Ans: (a) — GBAS CAT I: **±16 m horizontal, ±6 m vertical (95%)**.

Q225. GPS constellation: total operational SVs:

(a) 21

(b) 24

(c) 27

(d) 30

Ans: (b) — GPS: **24 SVs** (operational); up to 32 with spares.

Q226. GALILEO constellation: total SVs:

(a) 24

(b) 27

(c) 30

(d) 32

Ans: (c) — Galileo: **30 SVs**.

Q227. GLONASS constellation:

(a) 21

(b) 24

(c) 27

(d) 30

Ans: (b) — GLONASS: **24 SVs**.

Q228. GPS PRN codes — how many available?

(a) 24

(b) 32

(c) 64

(d) 252

Ans: (b) — GPS PRN codes: **32 available** (SVs numbered PRN 1–32).

Q229. GNSS integrity definition:

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(a) Accuracy of the position solution

(b) Ability to alert users when system should not be used

(c) Number of SVs visible

(d) Signal continuity

> **Ans: (b)** — Integrity: **ability to provide timely warning** when system unfit for navigation.

Q230. RAIM — what does it detect?

(a) Ionospheric error

(b) Failed SV causing position error exceeding limits

(c) Multipath

(d) Clock drift in receiver

> **Ans: (b)** — RAIM detects: **failed SV causing excessive position error**.

Q231. GPS almanac download time:

(a) 30 s

(b) 12.5 min

(c) 30 min

(d) 1 hour

> **Ans: (b)** — Full almanac download: **12.5 minutes** (25 frames × 30 s).

Q232. GPS cold start — time to first fix:

(a) 30 s

(b) 1–2 min

(c) 12.5 min

(d) Up to 30 min

> **Ans: (c)** — Cold start (no almanac): up to **12.5 min** for almanac download.

Q233. PDOP (Position Dilution of Precision) — best geometry:

(a) High PDOP

(b) Low PDOP

(c) PDOP = 1 always

(d) PDOP irrelevant

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Ans: (b) — Best geometry: **low PDOP** (SVs spread widely across sky).

Q234. GPS ionospheric model corrects approximately:

(a) 10%

(b) 50%

(c) 75%

(d) 99%

Ans: (b) — GPS ionospheric model: **≈50%** correction. Dual-freq: ≈99%.

Q235. SBAS integrity monitoring provided by:

(a) Ground reference stations

(b) Control segment only

(c) RAIM

(d) Aircraft receiver

Ans: (a) — SBAS: **network of ground reference stations** monitors satellite integrity.

Q236. Galileo orbital inclination:

(a) 55°

(b) 56°

(c) 63°

(d) 65°

Ans: (b) — Galileo: **56°**.

Q237. GPS Week Number rollover issue — period:

(a) 10.9 years (1024 weeks)

(b) 19.7 years

(c) 7 years

(d) Every year

Ans: (a) — GPS week: 10-bit number → rolls over every **1024 weeks (≈19.7 years)**. [Some say 19.7 — check ICAO doc for exact value used]

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Q238. GNSS continuity of service — definition:

(a) System accuracy

(b) Probability of system performing without interruption during operation

(c) Integrity

(d) Number of SVs

Ans: (b) — Continuity: **probability of uninterrupted operation during intended use**.

Q239. GNSS availability — definition:

(a) Accuracy spec

(b) Proportion of time system provides navigation with required accuracy

(c) Integrity

(d) Continuity

Ans: (b) — Availability: **proportion of time system meets accuracy and integrity requirements**.

Q240. GPS L1 C/A code — chip length:

(a) 30 m

(b) 300 m

(c) 1 km

(d) 3 km

Ans: (b) — C/A chip length = $c/f = 3\times10^8/1.023\times10^6$ = **293 m ≈ 300 m**.

Q241. GNSS approach — LPV (Localizer Performance with Vertical guidance) provides:

(a) Lateral only

(b) Lateral + baro-VNAV

(c) Lateral + SBAS vertical guidance (like ILS CAT I)

(d) No approach minima

Ans: (c) — LPV: SBAS approach with lateral + **SBAS vertical guidance**, minima approaching CAT I.

Q242. GPS time offset from UTC at receiver:

(a) Zero — GPS = UTC

(b) GPS ahead of UTC by integer seconds (leap seconds)

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(c) GPS behind UTC

(d) Varies daily

> **Ans: (b)** — GPS time: **ahead of UTC by integer leap seconds**. Receiver subtracts leap second count to display UTC.

Q243. GNSS — which error source causes errors of hundreds of metres without correction?

(a) Multipath

(b) Clock error

(c) Selective Availability (when active)

(d) Tropospheric delay

> **Ans: (c)** — When SA active: errors up to **100 m** (horizontal). SA now off.

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